

Plasma Modeling

Methods and Applications

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Plasma Modeling

Methods and Applications

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To Mariachiara, Gerardina, Michelangelo and Maria Teresa, our future.

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Preface

In the last few years, research activity in the fields of plasma physics and chemistry has grown exponentially, not only focusing on fundamental aspects, but also on industrial applications. Due to its inherent complexity, both experiments and numerical modeling are necessary to fully characterize plasma systems. The most commonly used numerical simulation techniques for plasma modeling include fluid dynamics, kinetic, and hybrid models. These simulation models differ significantly in terms of their principles, strengths, and limitations. Kinetic models, such as particle-in-cell with Monte Carlo collisions, are used for non-equilibrium systems, and fluid models are employed for faster computations, reducing the accuracy. Hybrid models provide balance between precision and efficiency.

The golden age of plasma modeling occurred in the 1980s–90s, when several research groups across the world developed theoretical and numerical approaches to investigate plasma behaviors. Currently, plasma modeling mainly addresses technological applications and, due to the complexity of systems, commercial multi-purpose codes are often used. These numerical codes are based on the fluid dynamics approach, describing realistic discharge geometries. *COMSOL Multiphysics* and *ANSYS Fluent* are examples of very popular commercial general-purpose software platforms, based on advanced numerical methods, for modeling and simulating physics-based problems. They are suitable for the design and development of optimized products by researchers and engineers working for leading technical enterprises, research laboratories, and universities. In strong non-equilibrium cases it is necessary to calculate the energy distribution function of particles. Kinetic modeling is recurrent in gas discharge investigations due to the non-equilibrium energy distribution of charged particles, which are not representable by the Maxwell distribution function, with abrupt changes in the slope, dumps and bumps, affecting the macroscopic properties of the plasma. In this context, a critical aspect is to determine the actual distribution function of free electrons, solving the Boltzmann equation. The electron energy distribution directly affects the plasma composition, also inducing non-equilibrium in internal states which, in turn, influences the electron gas, creating a strong synergy between the different plasma components. *BOLSIG+*, originally developed in the LAPLACE laboratory in Toulouse, is an example a freeware non-commercial code for the numerical solution of the Boltzmann equation for free electrons in weakly ionized gases in the framework of the two-term approximation.

Several non-commercial codes have been developed by research groups to simulate one- or two-dimensional problems and suitable plasma applications. It is worth mentioning *nonPDPSIM*, developed by the Kushner research group, which is a two-dimensional computational platform that solves, using the hybrid approach, transport equations for charged and neutral species, Poisson's equation for the electric potential, the electron energy conservation equation for the electron temperature, radiation transport and Navier–Stokes equations for the neutral gas flow, coupled with a Monte Carlo solver for the electron energy distribution function. The two-dimensional platform has three major modules addressing plasma dynamics,

fluid dynamics, and surface kinetics, and is implemented on an unstructured mesh to enable resolution of a large dynamic range in space.

The aim of this book is to present a review of the different approaches that can be adopted to model a plasma, giving a comprehensive bibliography for students and researchers who want to start a project in plasma modeling, orienting the reader toward the choice of a suitable numerical method. The chapters in this book have been written by experienced researchers in the field, who have developed advanced theoretical models and high-performance numerical codes. The theoretical models used in commercial or freeware codes are described throughout this book, guiding students, researchers, and engineers in choosing the proper approach for the problem under investigation. This book describes the theoretical and numerical approaches that the plasma community currently uses to investigate plasmas, and thus can help the reader in creating a customized in-house code, particularly when commercial or freeware codes cannot be used.

The book is divided into three parts: part 1, Kinetic theory; part 2, Fluid and hybrid models; and part 3, Applications. In part 1, the first chapter provides a general overview of the Boltzmann equation, methods of solution, and applications. The second and third chapters present the solution of the Boltzmann equation for free electrons in two-term and multi-term approximations. Chapter 4 describes the particle-in-cell algorithm. Chapter 5 presents the ergodic method, which is suitable for multi-scale problems in plasma physics.

In part 2, chapter 6 develops the theory of the fluid model for a plasma, while chapter 7 focuses on magnetohydrodynamics. Chapter 8 describes the self-consistent model, which couples state-to-state kinetics with free electron kinetics. Chapter 9 discusses hybrid models, considering electrons as a fluid and following ions as particles.

Finally, some applications are presented in part 3. In particular, chapter 10 shows the results for conventional and innovative plasma torches under local thermodynamic equilibrium, while chapter 11 focuses on non-equilibrium in high-pressure discharges, also considering the problem of fast gas heating. Chapter 12 discusses the role of radiation heating for different space vehicles entering planetary atmospheres. Chapter 13 treats the complex problem of dust formation in low pressure plasmas and chapter 14 examines the problem of the validation and verification of numerical codes, a fundamental aspect for modeling activities.

In each chapter, the authors have tried to find the right compromise in synthesizing a subject that would be worthy of its own book. Due to limitations of space, the editors were not able to dedicate discussion to some theoretical approaches, such as Monte Carlo and molecular dynamics, and some applications, such as waves in plasmas, self-organized plasmas, probe theory, and so on. Nevertheless, the theoretical and numerical approaches described in the book can be applied to any specific application. The extended reference section in each chapter will help in finding additional material for more in-depth analysis.

Gianpiero Colonna and Antonio D'Angola

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Part I

Kinetic Theory

Chapter 1

Boltzmann and Vlasov equations in plasma physics

If you ask researchers in physics and chemistry what the scientific discovery at the origin of modern science is, most of the answers will be Einstein's relativity or quantum physics. In our opinion, this role should be assigned to the *Boltzmann kinetic equation* and its application to the kinetic theory of gases. The same opinion was expressed by Max Planck in his Nobel lecture [1]:

... my trying to give you the story of the origin of the quantum theory in broad outlines and to couple with this, a picture in a small frame, of the development of this theory up to now, and its present-day significance for physics. ... For this reason, I busied myself, from then on, that is, from the day of its establishment, with the task of elucidating a true physical character for the formula, and this problem led me automatically to a consideration of the connection between entropy and probability, that is, Boltzmann's trend of ideas; until after some weeks of the most strenuous work of my life, light came into the darkness, and a new undreamed-of perspective opened up before me.

The Boltzmann equation introduced into physics the idea of probability, which was then used some years later in quantum physics. Also the concept of time changed, relating its direction to that of entropy exchange [2], breaking the symmetry between the past and the future [3], even if the laws of mechanics are invariant for time inversion. Despite its importance, students in physics and chemistry often confuse the Boltzmann equation with the Boltzmann distribution.

The Boltzmann equation has many applications in modern science [4, 5], from fluid dynamics to material science:

- The Chapman–Enskog expansion is a method for calculating the transport properties of gases [6].

- Numerical methods, such as the Bhatnagar–Gross–Krook (BGK) [7] approximation or *lattice Boltzmann method* (LBM) [8, 9], are commonly used to determine the flow field of complex fluids.
- The spectroscopic signature of gases as well as radiative heating are estimated by solving the Boltzmann equation for photons (see chapter 12).
- The thermal and electrical properties of metals and semiconductor materials can be estimated by solving the Boltzmann equation for electrons and phonons, adapted to account for quantum statistics [10].
- Mathematical aspects, such as the existence and unicity of solutions and the accuracy of approximate methods are also determined using the Boltzmann equation [11].

The Boltzmann equation is one of the most powerful tools for investigating the plasma state, from the electron kinetics in weakly ionized gases [12] to fusion [13] and astrophysical [14, 15] plasmas. In this chapter we present the theoretical foundation of the Boltzmann and Vlasov equations, giving an overview of their applications.

1.1 Fundamentals

Let us consider the single-particle phase space Φ , characterized by the sextuplet $(\mathbf{r}, \mathbf{v})$ of particle position and velocity. The state of the particle corresponds to a single point in this space, while a system with $\mathcal{N}$ particles is represented by a set of $\mathcal{N}$ points. In order to provide a pictorial view, we limit ourselves to a one-dimensional (1D) space coordinate and one velocity component. This representation can also be considered as the projection of the entire volume in the two-dimensional (2D) plane (x, v_x) . To determine the time evolution of the volume in the Φ space, the motion of each particle, according to the classical mechanic laws, must be followed:

$$\begin{aligned}\frac{d\mathbf{r}_i}{dt} &= \mathbf{v}_i \\ \frac{d\mathbf{v}_i}{dt} &= \mathbf{a}_i.\end{aligned}\tag{1.1}$$

The ensemble of points changes its shape, as can be observed in figure 1.1 (left) for a 2D example, because each particle moves independently of the others. The acceleration $\mathbf{a}_i$ originates from two contributions

$$\mathbf{a}_i = \mathbf{a}_{\text{ext}}(\mathbf{r}_i) + \sum_{k \neq i}^{\mathcal{N}} \frac{\mathbf{F}_{ik}}{m_i},\tag{1.2}$$

one, $\mathbf{a}_{\text{ext}}$, is due to an external field, and the other, $\mathbf{F}_{ik}/m_i$ where m_i is the particle mass, is due to intermolecular forces $\mathbf{F}_{ik}$. This approach, successfully applied to predict the orbit of planets in the solar system, in principle allows one to determine exactly the evolution of any N -particle system, such as a matter in the gas phase. In practice, the number of molecules in a macroscopic system ($\mathcal{N} \approx 10^{23}$) and the

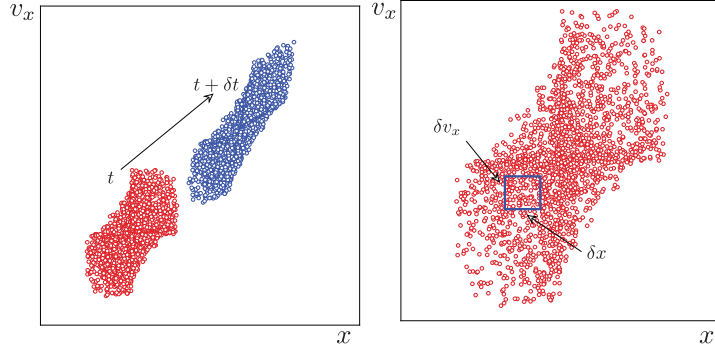


Figure 1.1. Projection of the phase space $(\mathbf{r}, \mathbf{v})$ on the 2D plane (x, v_x) of an N -particle system. Evolution of the sampling points with $a_x(x) = a_{x0} + a_{x1}x^2$ (left) and the starting distribution with the projection of a sampling volume $(\delta x, \delta v_x)$ (right).

uncertainty in the initial conditions¹ make such a deterministic procedure unfeasible for any numerical simulation, even for modern high-performance computers.

An alternative approach is to describe the evolution of an N -particle system by using the laws of statistics, developing the theory of *physical kinetics*. Let us consider the following assumptions:

1. A infinitesimal sampling volume, such as that depicted in figure 1.1 (right), contains a very large number of particles, so that a *macroscopic* infinitesimal volume still contains a large number of particles: the gas can be considered as a continuous medium.
2. The mean particle distance, $\bar{d} = \sqrt[3]{\frac{V}{N}}$, with V the physical volume, is larger than the thermal de Broglie length, $\lambda_{th,i} = \sqrt{\frac{2\pi\hbar}{m_i kT}}$: quantum effects are negligible and the classical statistics and laws of motion can be used.
3. The interparticle interaction distance $\bar{r}_i$ is much smaller than the mean-free-path λ_{mfp} : particle interactions can be described as binary collisions.

As a first step, let us limit ourselves to a system of free particles moving in an external field, without collisions. In a infinitesimal hypercube identified by the vertices $(\mathbf{r} - \frac{1}{2}\delta\mathbf{r}, \mathbf{v} - \frac{1}{2}\delta\mathbf{v})$ and $(\mathbf{r} + \frac{1}{2}\delta\mathbf{r}, \mathbf{v} + \frac{1}{2}\delta\mathbf{v})$, see figure 1.1 (right), which has a volume in the phase space of

$$\delta\Gamma = \delta x \delta y \delta z \delta v_x \delta v_y \delta v_z = \delta V \delta^3 v, \quad (1.3)$$

where δV is the physical volume, the number of particles in the infinitesimal hypercube can be written as

¹ The uncertainty of initial position and velocity is assured by *Heisenberg's principle*.

$$\delta\mathcal{N} = f(\mathbf{r}, \mathbf{v}, t)\delta\Gamma, \quad (1.4)$$

which defines the *distribution function* as the number of particles per unit volume in the phase space. For real systems, n_s different species can be present, each one with its own internal structure which, in the framework of quantum physics, can be represented by discrete levels, characterized by a set of quantum numbers ℓ .² This consideration brings us to a generalized definition of the distribution function, considering equation (1.4) for particles of the s th species in the ℓ th level, i.e.

$$\delta\mathcal{N}_{s,\ell}(t) = f_s(\mathbf{r}, \mathbf{v}, \ell, t)\delta\Gamma. \quad (1.5)$$

1.1.1 The convection operator

Let us now evaluate the rate at which the number of particles inside the sampling volume changes. It corresponds to the flux of particles crossing the surface of the hypercube. For the sake of simplicity, the discussion is restricted to the 2D problem depicted in figure 1.1 (right), with the square sampling volume, where the particle flux is the sum of four contributions

$$\frac{\partial\delta\mathcal{N}_{s,\ell}}{\partial t} = \Phi = \Phi_{\text{left}} + \Phi_{\text{right}} + \Phi_{\text{bottom}} + \Phi_{\text{top}}, \quad (1.6)$$

two through the vertical sides ('left' and 'right'), due to particles with the fixed velocity changing position

$$\begin{aligned} \Phi_{\text{left}} &= + \int_{v_x - \frac{1}{2}\delta v_x}^{v_x + \frac{1}{2}\delta v_x} w_x f_s\left(x - \frac{1}{2}\delta x, w_x, \ell, t\right) dw_x \\ &\approx v_x f_s\left(x - \frac{1}{2}\delta x, v_x, \ell, t\right) \delta v_x \\ \Phi_{\text{right}} &= - \int_{v_x - \frac{1}{2}\delta v_x}^{v_x + \frac{1}{2}\delta v_x} w_x f_s\left(x + \frac{1}{2}\delta x, w_x, \ell, t\right) dw_x \\ &\approx -v_x f_s\left(x + \frac{1}{2}\delta x, v_x, \ell, t\right) \delta v_x, \end{aligned} \quad (1.7)$$

and two through the horizontal sides ('bottom' and 'top'), due to particles accelerated in the same positions

² As an example, for atomic species we have the quadruplet (n, l, m, s) corresponding to the principal, angular, magnetic, and spin quantum numbers, and for molecules (n, J, v, s) corresponding to the electronic state, and rotational, vibrational, and total spin quantum numbers. However, internal levels can be ordered by energy and identified by a single integer number.

$$\begin{aligned}
 \Phi_{\text{bottom}} &= + \int_{x-\frac{1}{2}\delta x}^{x+\frac{1}{2}\delta x} a_x(x') f_s\left(x', v_x - \frac{1}{2}\delta v_x, \ell, t\right) dx' \\
 &\approx a_x(x) f_s\left(x, v_x - \frac{1}{2}\delta v_x, \ell, t\right) \delta x \\
 \Phi_{\text{top}} &= - \int_{x-\frac{1}{2}\delta x}^{x+\frac{1}{2}\delta x} a_x(x') f_s\left(x', v_x + \frac{1}{2}\delta v_x, \ell, t\right) dx' \\
 &\approx -a_x(x) f_s\left(x, v_x + \frac{1}{2}\delta v_x, \ell, t\right) \delta x.
 \end{aligned} \tag{1.8}$$

Combining the terms in equations (1.7) and (1.8), we obtain

$$\begin{aligned}
 \Phi_{\text{left}} + \Phi_{\text{right}} &= v_x \delta v_x \left[f_s\left(x - \frac{1}{2}\delta x, v_x, \ell, t\right) - f_s\left(x + \frac{1}{2}\delta x, v_x, \ell, t\right) \right] \\
 \Phi_{\text{bottom}} + \Phi_{\text{top}} &= a_x(x) \delta x \left[f_s\left(x, v_x - \frac{1}{2}\delta v_x, \ell, t\right) - f_s\left(x, v_x + \frac{1}{2}\delta v_x, \ell, t\right) \right].
 \end{aligned} \tag{1.9}$$

Considering that the sampling volume in the phase space is small, the parentheses on the right-hand side in equation (1.9) can be approximated with the derivatives

$$\begin{aligned}
 f_s\left(x - \frac{1}{2}\delta x, v_x, \ell, t\right) - f_s\left(x + \frac{1}{2}\delta x, v_x, \ell, t\right) &\approx -\frac{\partial f_s(x, v_x, \ell, t)}{\partial x} \delta x \\
 f_s\left(x, v_x - \frac{1}{2}\delta v_x, \ell, t\right) - f_s\left(x, v_x + \frac{1}{2}\delta v_x, \ell, t\right) &\approx -\frac{\partial f_s(x, v_x, \ell, t)}{\partial v_x} \delta v_x
 \end{aligned} \tag{1.10}$$

and equations (1.6)–(1.8) give the total flux as

$$\frac{\partial \delta \mathcal{N}_{s,\ell}}{\partial t} = \Phi = -\delta x \delta v_x \left[v_x \frac{\partial f_s(x, v_x, \ell, t)}{\partial x} + a_x(x) \frac{\partial f_s(x, v_x, \ell, t)}{\partial v_x} \right]. \tag{1.11}$$

To extend this result to the whole phase space, we should note that in multi-dimensional space:

- The integrals in equations (1.7) and (1.8) must be performed over all the variables apart from the one perpendicular to the surface, i.e. $\delta x \delta v_x \rightarrow \delta \Gamma$.
- Two contributions must be considered for each independent direction, i.e. equation (1.9) should also be written for the y - and z -directions.
- Acceleration depends on the position in the phase space, i.e. $a_x(x) \rightarrow [a_x(\mathbf{r}), a_y(\mathbf{r}), a_z(\mathbf{r})]$ ³

³ The acceleration can also depend on the velocity, as happens in the presence of a magnetic field for charged species, an aspect discussed in chapter 2.

These assumptions, together with equation (1.5), give

$$\frac{\partial \delta \mathcal{N}_{s,\ell}}{\partial t} = \delta \Gamma \frac{\partial f_s(\mathbf{r}, \mathbf{v}, \ell, t)}{\partial t} = -\delta \Gamma [\mathbf{v} \cdot \nabla_r + \mathbf{a}(\mathbf{r}) \cdot \nabla_v] f_s(\mathbf{r}, \mathbf{v}, \ell, t), \quad (1.12)$$

where

$$\begin{aligned} \mathbf{v} \cdot \nabla_r &\equiv v_x \frac{\partial}{\partial x} + v_y \frac{\partial}{\partial y} + v_z \frac{\partial}{\partial z} \\ \mathbf{a}(\mathbf{r}) \cdot \nabla_v &\equiv a_x(\mathbf{r}) \frac{\partial}{\partial v_x} + a_y(\mathbf{r}) \frac{\partial}{\partial v_y} + a_z(\mathbf{r}) \frac{\partial}{\partial v_z}. \end{aligned} \quad (1.13)$$

The square parentheses in equation (1.12) can be considered as a convection operator, because they describes the evolution of the distribution function due to the free motion of particles subjected to an external force.

1.1.2 The collisional operator

Let us now describe the contribution of collisions to the evolution of the distribution in the phase space. If the gas density is sufficiently small only binary encounters can be considered. If the mean free path is much larger than the interaction distance, these encounters can be treated as collisions. In a collision at given particle velocities, the final state is completely determined by the deflection angle and, in the case of internal excitation and chemical reactions, by the energy necessary to activate the considered process. The uncertainty in assigning position and speed to the colliding particles makes it necessary to describe the collisions statistically. The cross section of a process is the measure of the probability of a collision output. Let us define the following quantities:

- The differential cross section $\frac{d\sigma}{d\Omega}$ is the ratio between the number of particles deflected in the unit time in a given solid angle $d\Omega$ and the incoming particle flux per unit time and per unit surface. It is a function of the modulus of the relative velocity $|\mathbf{v}_A - \mathbf{v}_B|$, the initial and final levels and the solid angle.
- The total cross section σ is the integral of the differential cross section over the whole solid angle. For elastic collisions this quantity is not relevant, including also the contribution of collisions with very small deflection angles. On the other hand, it is often used in inelastic collisions, whose angular dependence on differential cross section is assumed to be isotropic.

Including the collisional operator, equation (1.14) becomes the Boltzmann equation in the usual form

$$\frac{\partial f_s(\mathbf{r}, \mathbf{v}, \ell, t)}{\partial t} + [\mathbf{v} \cdot \nabla_r + \mathbf{a}(\mathbf{r}) \cdot \nabla_v] f_s(\mathbf{r}, \mathbf{v}, \ell, t) = \left(\frac{\delta f_s}{\delta t} \right)_{\text{coll}}. \quad (1.14)$$

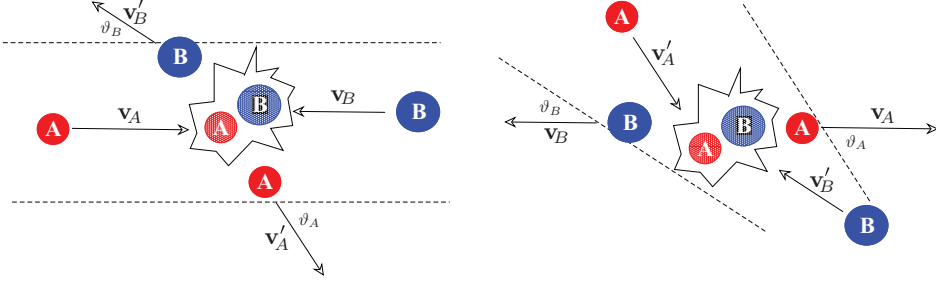


Figure 1.2. Scheme of a binary elastic collision (left) with the deflection angle and a picture of the reverse process (right) obtained by reversing time and space. The dashed lines are the directions of the velocities before the collisions and θ are the corresponding deflection angles.

Let us construct the collisional operator analyzing the binary encounter sketched in figure 1.2. In a sampling volume given by equation (1.3), the number of particles A with velocity $\mathbf{v}_A$ flowing through a unitary surfaces moving jointly with B is given by⁴

$$\Phi_A(\mathbf{v}_A, \mathbf{v}_B) = f_A |\mathbf{v}_A - \mathbf{v}_B| \delta^3 v_A = f_A v_{AB} \delta^3 v_A, \quad (1.15)$$

where $v_{AB} = |\mathbf{v}_A - \mathbf{v}_B|$ is the relative velocity and, from the definition of the differential cross section,⁵ the number of particles A deflected by the angle ϑ_A in the unit time by a single particle B (see figure 1.3) is

$$\begin{aligned} \left[\frac{d\delta \mathcal{N}_A(\vartheta_A)}{dt} \right]_1 &= \Phi_A(\mathbf{v}_A, \mathbf{v}_B) \frac{d\sigma_{AB}}{d\Omega}(\vartheta_A) \delta\Omega_A \\ &= f_A v_{AB} \delta^3 v_A \frac{d\sigma_{AB}}{d\Omega}(\vartheta_A) d\Omega_A, \end{aligned} \quad (1.16)$$

with

$$d\Omega_A = \sin \vartheta_A d\vartheta_A d\phi, \quad (1.17)$$

with ϕ the angle of the deflection plane.⁶ The velocities after the collision $\mathbf{v}'_A$ and $\mathbf{v}'_B$, as well as ϑ_B , are univocally determined by ϑ_A and ϕ , because energy and momentum conservation laws can be used to determine all the other components. Equation (1.16) gives the number of particles undergoing the considered collision with a single B particle. To have the total number of particles deflected in the given direction we have to multiply this quantity by the number of scattering centers, i.e.

$$\frac{\partial \delta \mathcal{N}_A(\vartheta_A)}{\partial t} = \left[\frac{\partial \delta \mathcal{N}_A(\vartheta_A)}{\partial t} \right]_1 f_B \delta^3 v_B \delta V = f_A f_B v_{AB} \frac{d\sigma_{AB}}{d\Omega}(\vartheta_A) \delta^3 v_A \delta^3 v_B \delta V d\Omega_A. \quad (1.18)$$

⁴ For sake of simplicity, let us consider implicitly the variables of the distribution substituting $f_{A/B}(\mathbf{r}, \mathbf{v}_{A/B}, \ell_{A/B}, t) \rightarrow f_{A/B}$ and $f_{A/B}(\mathbf{r}, \mathbf{v}'_{A/B}, \ell'_{A/B}, t) \rightarrow f'_{A/B}$.

⁵ Let us write explicitly only the dependence of the cross section on the deflection angle, using the symbol $\frac{d\sigma'}{d\Omega}$ for the reverse process.

⁶ The directions of the velocities after the collision, if not parallel, univocally define the deflection plane. Therefore the same angle ϕ can be used for both particles.

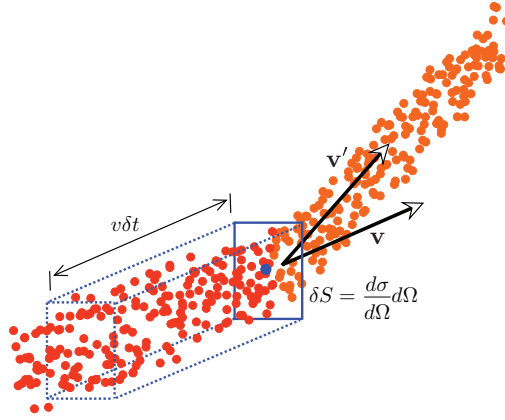


Figure 1.3. The number of particles, all with the same velocity $\mathbf{v}$ relative to the scattering center (blue dot), flowing through the surface δS in the time interval δt are those contained in the volume $\delta V = v \delta t \delta S$. Considering δS as the differential cross section, all these particles are deflected in the same solid angle $d\Omega$ with velocity $\mathbf{v}'$.

It should be noted that equation (1.18) must be symmetric with respect to exchange between A and B , because this expression gives the same rate for the number of particles B deflected by the angle θ_B , therefore

$$\frac{d\sigma_{AB}}{d\Omega}(\vartheta_A)d\Omega_A = \frac{d\sigma_{BA}}{d\Omega}(\vartheta_B)d\Omega_B. \quad (1.19)$$

The variation expressed by equation (1.18) is the loss rate for particles A and B in the state before the collision. At the same time, the gain rates coming from the reverse process must be considered, as illustrated in figure 1.2 (right), given by

$$\left[\frac{\partial \delta \mathcal{N}_A(\vartheta_A)}{\partial t} \right]_r = f'_A f'_B v'_{AB} \delta^3 v'_A \delta^3 v'_B \frac{d\sigma'_{AB}}{d\Omega}(\vartheta_A) d\Omega_A \delta V, \quad (1.20)$$

noting that the deflection angles are the same as in the direct process. However, to maintain parity with respect to space–time inversion, the probability of a process must be equal to that of its reverse, i.e.

$$v'_{AB} \frac{d\sigma'_{AB}}{d\Omega}(\vartheta_A) = v_{AB} \frac{d\sigma_{AB}}{d\Omega}(\vartheta_A). \quad (1.21)$$

This equation is known as a *micro-reversibility* principle. This assumption can be demonstrated for collisions with spherical symmetry but it cannot be proved in general. For strange geometries, such as those of large organic molecules, the micro-reversibility principle is not valid. Nonetheless, in this book the micro-reversibility principle is considered as valid. The micro-reversibility principle has a corresponding law from the macroscopic point of view, the *detailed balance* principle. The global variation of the distribution function of f_A is obtained by summing over all the collision partners B , integrating over all its initial states and over all the final states of A , thus obtaining the final expression for the collisional operator

$$\left(\frac{df_A}{dt}\right)_{\text{coll}} = \sum_{\ell'_A} \sum_{B, \ell_B, \ell'_B} \int v_{AB} \frac{d\sigma_{AB}}{d\Omega}(\vartheta_A) (f'_A f'_B - f_A f_B) d^3v_B d\Omega_A, \quad (1.22)$$

that has been divided by the volume $\delta\Gamma_A = \delta V \delta^3v_A$ to obtain the equation for the distribution function.

1.1.3 Boltzmann's H -theorem

It is straightforward to demonstrate that in equilibrium, i.e. in the presence of Boltzmann and Maxwell distributions, at the same temperature, the collisional operator is null, reproducing an important result already obtained in the framework of statistical physics. Another important result, obtained for the first time by Boltzmann, is the demonstration of the growth of entropy for an isolated rarefied gas, namely, the H -theorem. Let us define the functional

$$H(t) = \iiint d^3v f \ln f \quad (1.23)$$

and its time derivative⁷ given by

$$\frac{\partial H(t)}{\partial t} = \int d^3v \frac{\partial f}{\partial t} (\ln f + 1). \quad (1.24)$$

Substituting the Boltzmann equation (see equation (1.14)) in this expression and noting that the convection operator in equation (1.12) gives a null integral⁸, only the contribution of the collision integral remains. Distinguishing the colliders as in the previous section we obtain

$$\frac{\partial H(t)}{\partial t} = \int d^3v_A (\ln f_A + 1) v_{AB} \frac{d\sigma_{AB}}{d\Omega}(\vartheta_A) (f'_A f'_B - f_A f_B) d^3v_A d^3v_B d\Omega_A. \quad (1.25)$$

The argument of the integral in equation (1.25) is invariant by exchanging A and B and averaging the two expressions we obtain

$$\frac{\partial H(t)}{\partial t} = \frac{1}{2} \int d^3v_A [\ln(f_A f_B) + 2] v_{AB} \frac{d\sigma_{AB}}{d\Omega}(\vartheta_A) (f'_A f'_B - f_A f_B) d^3v_A d^3v_B d\Omega_A. \quad (1.26)$$

As a consequence of the micro-reversibility principle, the integral must be the same when the initial and final states are exchanged

$$\frac{\partial H(t)}{\partial t} = \frac{1}{2} \int d^3v_A [\ln(f'_A f'_B) + 2] v_{AB} \frac{d\sigma_{AB}}{d\Omega}(\vartheta_A) (f_A f_B - f'_A f'_B) d^3v_A d^3v_B d\Omega_A, \quad (1.27)$$

⁷ It should be noted [16, 17] that the entropy is defined as the integral of $(f - f \ln f)$. The following demonstration can be applied in a straightforward manner to entropy [16], showing that the time derivative of S has the opposite sign with respect to that of H .

⁸ This assertion can be easily demonstrated, by applying Gauss's theorem and transforming the volume integral of the derivatives in a surface integral of the function. At infinity the distribution function is null, therefore the surface integral gives a null contribution [16].

and averaging equations (1.26) and (1.27) we have the final expression

$$\begin{aligned} \frac{\partial H(t)}{\partial t} &= \frac{1}{4} \int d^3v_A [\ln(f_A f_B) - \ln(f'_A f'_B)] v_{AB} \\ &\times \frac{d\sigma_{AB}}{d\Omega} (\partial_A) (f'_A f'_B - f_A f_B) d^3v_A d^3v_B d\Omega_A. \end{aligned} \quad (1.28)$$

The sign of the integrand depends on the product

$$[\ln(f_A f_B) - \ln(f'_A f'_B)] (f'_A f'_B - f_A f_B), \quad (1.29)$$

which is always negative or null when the initial and final distributions are the same, conditions verified at the equilibrium. The integral of a negative function is always negative, it is

$$\frac{\partial H(t)}{\partial t} \leq 0, \quad (1.30)$$

demonstrating the H -theorem.⁹

1.1.4 The Vlasov equation

The above theory cannot be applied for ionized gases, as they involve the Coulomb interaction, a long-range force. As a consequence charge–charge interaction cannot be treated as a collision. If the gas is weakly ionized, the charge–charge interactions are negligible with respect to collisions with neutrals, but for highly ionized gases the Coulomb forces provide the dominant contribution, and some adjustments are necessary for the Boltzmann equation to be applied to plasmas.

Let us consider a test particle s with charge q_s , position $\mathbf{r}_s$ in a plasma in stationary conditions. The s particle will feel the electric field produced by all the other charges

$$\mathbf{E}(\mathbf{r}_s, t) = \frac{1}{4\pi\epsilon_0} \sum_{i \neq s}^N \frac{q_i \mathbf{r}_{s,i}(t)}{|\mathbf{r}_{s,i}(t)|^3}, \quad (1.31)$$

where $\mathbf{r}_{s,i} = \mathbf{r}_i(t) - \mathbf{r}_s$ is the position of i relative to s . We can separate two contributions to $\mathbf{E}(\mathbf{r}_s, t)$, a constant mean field

$$\mathbf{E}_0(\mathbf{r}_s) = \frac{1}{\delta t} \int_t^{t+\delta t} \mathbf{E}(\mathbf{r}_s, t) dt, \quad (1.32)$$

where δt is a time interval sufficiently long, and $\mathbf{E}_r(\mathbf{r}_s, t)$, due to fluctuations, is a function of time with null mean

$$\frac{1}{\delta t} \int_t^{t+\delta t} \mathbf{E}_r(\mathbf{r}_s, t) dt = \frac{1}{\delta t} \int_t^{t+\delta t} [\mathbf{E}(\mathbf{r}_s, t) - \mathbf{E}_0(\mathbf{r}_s)] dt \approx \mathbf{0}. \quad (1.33)$$

⁹ The demonstration given here refers to homogeneous systems. For non-homogeneous systems see [5].

The time interval δt must be a macroscopic time, i.e. $\delta t \gg 1/\tau_c$ where τ_c is the mean time between collisions. In principle, $\mathbf{E}_r(\mathbf{r}_s, t)$ is a continuous function of time, with rapid oscillations with mean time amplitude of the order of τ_c . This means that if we sample this field periodically, at successive equal time intervals of amplitude δt , much larger than τ_c , i.e. $\mathbf{E}_r(\mathbf{r}_s, t)$, $\mathbf{E}_r(\mathbf{r}_s, t + 2\delta t)$, ..., $\mathbf{E}_r(\mathbf{r}_s, t + n\delta t)$, we will have a random series. If the system is not in a stationary condition, the above assumptions remain valid if the sampling time δt is small with the respect to the evolution time of whole system, i.e. all the macroscopic quantities are almost constant in this time interval. Similar arguments can be used for the magnetic field. This property of the local field suggests that the mean field contributes to the acceleration in the Boltzmann equation, while $\mathbf{E}_r$ must contribute to the collision integral, due its random evolution. The mean field does not produce entropy, a consequence of the H -theorem, differently from the random field.

In fully ionized gases it could be demonstrated that the entropy variation due to collisions is much slower than other macroscopic quantities such as charge density ρ_c ,

$$\left| \frac{1}{S} \frac{\partial S}{\partial t} \right| \ll \left| \frac{1}{\rho_c} \frac{\partial \rho_c}{\partial t} \right|, \quad (1.34)$$

therefore the collision integral can be considered null. The Boltzmann equation in equation (1.14) becomes

$$\frac{\partial f_s}{\partial t} + \left[\mathbf{v} \cdot \nabla_r + \frac{q_s}{m_s} (\mathbf{E}_0 + \mathbf{v} \wedge \mathbf{B}_0) \cdot \nabla_v \right] f_s = 0, \quad (1.35)$$

where the local fields are related only to local macroscopic quantities such as charge density and current,

$$\begin{cases} \rho_c = \sum_s q_s \int f_s d^3v \\ \mathbf{j}_c = \sum_s q_s \int \mathbf{v} f_s d^3v, \end{cases} \quad (1.36)$$

through the Maxwell equations

$$\begin{cases} \nabla \cdot \mathbf{E}_0 = \frac{\rho_c}{\epsilon_0} \\ \nabla \cdot \mathbf{B}_0 = 0 \\ \nabla \wedge \mathbf{E}_0 = -\frac{\partial \mathbf{B}_0}{\partial t} \\ \nabla \wedge \mathbf{B}_0 = \frac{\mathbf{j}_c}{\epsilon_0 c^2} + \frac{1}{c^2} \frac{\partial E_0}{\partial t}. \end{cases} \quad (1.37)$$

This system is known as the *Vlasov equation*. In most practical applications the time derivatives of the fields in the Maxwell equations are negligible, and the quasi-static fields are considered. The validity of the Vlasov equation depends on two conditions:

- The mean particle distance must be smaller than the Debye length

$$\lambda_D = \sqrt{\frac{\epsilon_0 k T V}{\sum_s q_s^2 \mathcal{N}_s}}, \quad (1.38)$$

i.e. the number of particles inside the Debye sphere must be statistically significant [16].

- The collision frequency is much smaller than the plasma frequency

$$\omega_p = \sqrt{\sum_s \frac{q_s^2 \mathcal{N}_s}{\epsilon_0 m_s V}} \quad (1.39)$$

in order to neglect the collision integral.

1.2 Cross sections

The collisional term in the Boltzmann equation results from the dynamics of binary collisions in the plasma promoting elastic, inelastic, and reactive processes that govern the temporal evolution of the kinetics at a microscopic level. The construction of a reliable database for the elementary probabilities represents a challenge even today, due to the requirements of accuracy, consistency, and completeness, issues that significantly affect the predictive capability of kinetic models, not only in the estimation of macroscopic plasma parameters, but also in shedding light on relevant collisional mechanisms [18]. Modern approaches in quantum molecular dynamics are fully exploited to aim for the highest accuracy. However, the wide spectrum of theoretical methods, including semiclassical, quasi-classical, classical, and semi-empirical, provides methods that are suitable for specific classes of processes, offering a favorable balance between the reliability of dynamics information and the computational load characterizing the derivation of a complete dataset. This is, in particular, the case for state-to-state kinetics (see section 8.1), where the dependence of the probability on the quantum state of colliders is also considered.

Elastic collisions usually dominate the kinetics and fully determine the transport properties of thermal equilibrium plasmas [6]. Transport cross sections, i.e. the momentum transfer and higher ℓ moments of an elastic cross section, and their thermal average, i.e. the collision integrals $\Omega_{ij}^{\ell,s}(T)$ of the Chapman–Enskog theory [19], can be obtained by quantum [20] or semiclassical Wentzel-Kramers-Brillouin (WKB) estimation of phase-shifts, or by direct classical trajectory integration of the interaction potential [21]. The interaction potential from accurate electronic structure calculations, or modeled with either full-range analytical forms [6] or a phenomenological potential [22], is assumed to be isotropic, so is dependent only on the approaching line of colliding partners, also for interactions involving molecules. Some examples do exist in the literature of classical trajectory calculations on

ab initio surfaces for molecular collisions [23], considering anisotropic potentials, however, for small rapidly rotating molecules an isotropic angle-averaged potential can be assumed [24]. Resonant charge exchange in ion–parent-atom interactions is the only non-elastic process considered in transport calculations, due to the significant contribution in the ℓ -odd collision integrals [6]. A rigorous quantum coupled-channel approach [20] is required; however, it is worth mentioning the asymptotic approach [25, 26], based on the idea of expanding the cross section in terms of small parameters connected to the geometry of the collision.

In the case of electron elastic scattering, the quantum nature of the interactions is not negligible and manifests itself in the features of the energy profile of the cross section, as the low-energy peaks due to resonances, or the *Ramsauer minimum* [27]. Therefore a quantum approach is required and the collision integrals are evaluated by integration of theoretical [28] or experimental differential elastic electron-scattering cross sections.

Inelastic and reactive channels are considered negligible in transport, except for quantum effects at low temperatures. However, they play a key role in non-equilibrium plasmas, chemical species undergoing excitation, and dissociation and ionization processes induced by electron impact or collisions with heavy particles with increasing threshold energies.

Quantum approaches (the R-matrix [29], close-coupling, the Schwinger multi-channel [30], local complex potential [31]) represent the best theoretical framework for the investigation of the dynamics in electron–molecule scattering. They involve non-adiabatic effects as the resonant electron capture with the formation of long-living negative ion intermediates or vibronic coupling. However, other methods have been successfully used in the past: first-order approaches, such as the Born, Born–Bethe, and Born–Ochkur approaches [32], which are accurate for high collision energies; semiclassical methods [33–35] that assume that the impinging electron moves classically in linear trajectories and the quantum molecular target; and the classical Gryzinskii method [36]. More recently, novel simplified approaches have been proposed in the literature, such as the BE*f*-scaling method [37], based on a rescaling procedure of the Born cross section for dipole-allowed vibronic excitations with the experimental optical oscillator strengths, and the similarity approach [38] or the binary encounter dipole (BED) model for ionization [39, 40]. Simplified approaches are still attractive, demonstrating in many cases striking agreement with experiments and representing very promising tools for more complex molecular systems.

For the dynamics of heavy-particle collisions, leading to energy transfer among internal and translation degrees of freedom of colliding partners, rearrangement reactions, or dissociation, the quasi-classical trajectory (QCT) method [18, 41, 42] has been well assessed. It has proved to compare well with and complement the quantum approaches [43]. The QCT method provides accurate potential energy surface (PES) values and robust trajectory statistics, allowing the derivation of state-specific rate coefficients through the pseudo-quantization of the energy in reactant and product channels, and is able to fully exploit the advantages of high-performance code parallelization [44]. Furthermore, the QCT approach can follow

the dynamics in collisional systems characterized by non-adiabatic coupling among different electronic states, with trajectories hopping [45] from ground to excited PESs. The mechanism of vibrational energy transfer in atom–molecule and molecule–molecule collisions has also been investigated in the framework of the semiclassical approach [46], the forced-harmonic oscillator [47], and the widely used SSH method [48].

A number of different open-access web-databases are available [49–52], where dynamical information has been collected, validated, and shared with the modeling community. These also include the state-specific cross sections and rate coefficients of elementary processes involving ground and excited chemical species, with resolution on the electronic, vibrational, and rotational degrees of freedom [53, 54].

1.3 Solution of the Boltzmann equation

The first approach for an analytical solution of the Boltzmann equation was the expansion series called *Chapman–Enskog* theory [55], which writes the distribution function as

$$f = f_0 + \sum_{n=1}^{\infty} \epsilon^n f_n, \quad (1.40)$$

where ϵ is a small parameter, and stops the series at the first order. The first term f_0 is the Maxwell distribution, which gives a null collision integral. Therefore the transport coefficients are related to the perturbation f_1 . This approach allows one to calculate the transport properties relative to the specific cross sections. Also, the perturbation is expanded in a polynomial series [6], allowing different degrees of approximation. The Chapman–Enskog theory is valid for small deviations from the Maxwell distributions. The Chapman–Enskog theory leads to the Navier–Stokes equations. However, in extremely rarefied gases, Navier–Stokes equations fail and more accurate approaches are required. An improved model was introduced by Grad [56] extending the moment equations to thirteen terms.

Even if it is very difficult solve the complete Boltzmann equation due to the large dimension of the phase space and the non-linearity of the collisional operator, different numerical methods have aspired to this. In recent years the lattice Boltzmann method (LBM) [8, 57] has been the subject of growing interest to simulate complex fluids. This method writes the Boltzmann equation in a discrete space $\mathcal{L}$ with a finite number of possible velocities (see figure 1.4). The time must be discrete as well in order to obtain that any advancement in the position is still a point of the lattice, i.e.

$$x_i \in \mathcal{L} \Rightarrow x_i + \mathbf{v}_j \delta t \in \mathcal{L}. \quad (1.41)$$

An interesting feature of the LBM is that it is a solver of the Navier–Stokes equations [8, 57], a characteristic inherited from the original Boltzmann equation. In contrast to the Navier–Stokes equations, the LBM has a linear convective operator. In most cases, the collisional operator is also linearized, using the BGK approach [9]. This method is practical and suitable for investigating complex systems, such as incompressible flows through porous media [58] and multi-phase/multi-component flows [59].

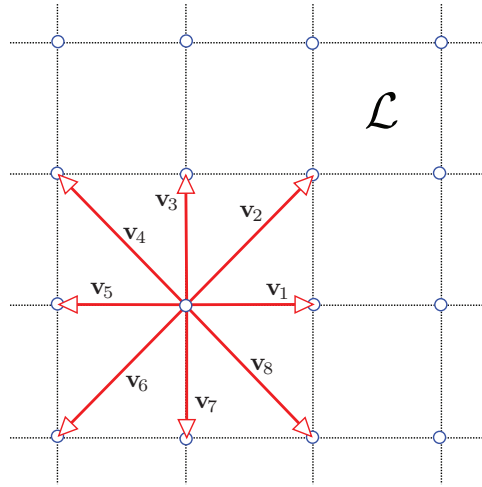


Figure 1.4. Example of a 2D lattice with discrete velocities.

Even if the LBM can be derived directly from the Boltzmann equation in the BGK approximation [7] under small Mach number expansion [60], the two equations are substantially different. Whereas the Boltzmann equation is valid for rarefied gases, the LBM is mainly used for incompressible flows and dense matter. Moreover, the H -theorem is not applicable in LBM.

Another interesting approach based on the Boltzmann equation is the *direct simulation Monte Carlo* (DSMC), developed by G Bird in the early 1960s [61, 62]. The method has been developed to investigate rarefied flows in very low density, where the Navier–Stokes equations fail, and is based on the idea that the free motion and the collisions can be decoupled. The DSMC considers the atomic structure of the matter. The gas is represented by a large number of pseudo-particles, each one representing a large number of real particles in the same position and with the same velocity. In the convection phase all the particles move according to the classical laws of motion for a given time interval, δt , which is small with respect to the mean collision time. Figure 1.1 (left) can also be considered a pictorial view of this system. The computational domain is divided into a grid (see figure 1.1 (right) for an example of grid elements) with the purpose of calculating statistical means, such as density, velocity, temperature, and so on. In each grid element, according to the total cross section, the number of collisions is determined and a Monte Carlo approach is used to sample the collisions. Random numbers are generated to decide which particles will collide and the result of the collision, changing the state of the particles accordingly. Convective and collision phases are repeated one after the other. With this method it is possible to calculate the distribution function directly, with a sensitivity depending on the number of particles. DSMC has been used to model hypersonic entry into a planetary atmosphere [63], and also for real spacecraft, such as the space shuttle [64, 65].

For fluid dynamics applications, simplified cross-section models are used, such as hard spheres [66], variable hard spheres [67], and variable soft spheres [68], since the

collision probabilities are available in closed form. More advanced models include cross sections obtained from quantum or semiclassical calculations (see section 1.2), also including state-to-state kinetics [69–71], as described in chapter 8. The inclusion of detailed cross-section datasets in DSMC calculations is a complex problem, not only due to the large amount of memory required to store the data and the time needed to calculate the probability of each event, but also because these data are not easily available. A novel approach, called DSMC-QCT [72, 73], does not go through the cross-section evaluation, but instead follows the motion of particles on a multi-dimensional PES. Once the colliding particles have been chosen, random numbers are generated to select the collisional parameters. The interaction is then followed in the center-of-mass frame of reference, moving the particles on the given PES, until the interparticle distance is large enough. The final state obtained in this way determines the new states of the colliders.

DSMC is not adequate for modeling plasma, although some attempts have been made in the literature using different time-steps for heavy particles and electrons [74]. Other approaches entail the coupling of particle-in-cell (PIC) (see chapter 4) with DSMC to model plasmas in rarefied flows [75].

The Boltzmann equation has also been applied to calculate the evolution of non-equilibrium distributions of electrons and phonons in metals heated by femtosecond laser pulses [10, 76–82]. The laser pulse is absorbed by the electron gas and the energy is redistributed by electron–electron collisions. At longer times the energy is transferred to the solid lattice with the production of phonons.

In a metal, electrons partially occupy the conduction band, following the Fermi statistic. In this case we generalize the definition of distribution as

$$n_s = \frac{\mathcal{N}_s}{V} = \int_0^\infty g_s f_s^{\text{FD}}(\varepsilon) \mathcal{D}_s(\varepsilon) d\varepsilon, \quad (1.42)$$

where g_s is the spin multiplicity of the particle s ($= 2$ for electrons) and $\mathcal{D}_s$ is the density of states. The equilibrium state is described by the Fermi–Dirac distribution

$$f_s^{\text{FD}}(\varepsilon) = (e^{\beta(\varepsilon - E_F)} + 1)^{-1}, \quad (1.43)$$

where E_F is the Fermi energy,¹⁰ defined as

$$\mathcal{N}_s = \int_0^{E_F} \mathcal{D}_s(\varepsilon) d\varepsilon. \quad (1.44)$$

Even at absolute temperature $T = 0$ the electrons are not in the same quantum state because of the Pauli exclusion principle, but all allowed states below the Fermi energy are fully occupied and the distribution function is a step function

$$\begin{aligned} \varepsilon \leq E_F &\Rightarrow f_s^{\text{FD}}(\varepsilon) = 1 \\ \varepsilon > E_F &\Rightarrow f_s^{\text{FD}}(\varepsilon) = 0. \end{aligned} \quad (1.45)$$

¹⁰Equation (1.43) is only an approximation and is valid only at low temperatures.

This character of the Fermi distribution should be reflected in the Boltzmann equation and, in particular, to account for the Pauli exclusion principle the product of the distributions in the collision integral (see equation (1.22)) becomes

$$f_A f_B \rightarrow f_A(1 - f_A) f_B(1 - f_B), \quad (1.46)$$

which assures that the distribution does not become larger than one.

1.4 Plasma modeling numerical codes

To conclude this chapter, we will give a brief description of the numerical codes commonly used by researchers for modeling plasmas, using the theoretical approaches described in the following chapters. The most used numerical simulation techniques for plasma modeling include fluid dynamics, kinetic, and hybrid models. These simulation models are significantly different in terms of their principles, strengths, and limitations. Kinetic models, such as PIC with Monte Carlo collisions (see chapter 4), are used for non-equilibrium systems, and fluid models are employed for faster computation with the cost of reducing the accuracy. Hybrid models provide balance between precision and efficiency.

COMSOL Multiphysics [83] is a general-purpose software platform, based on advanced numerical methods, for modeling and simulating physics-based problems. It is suitable for the design and development of optimized products by researchers and engineers working for leading technical enterprises, research laboratories, and universities. The *COMSOL* platform is characterized by an environment designed for cross-disciplinary product development with a unified workflow for electrical, mechanical, fluid, and chemical applications. The ‘Plasma’ module, available in the ‘Electrical’ suite, is a partial differential equation solver including drift–diffusion approximations for ions, the quasi-neutrality assumption for electron movements, reduced Maxwell equations for electromagnetic fields, electron energy equations for electron temperatures, and the Navier–Stokes equation for neutral background gas. Users can conveniently formulate physical equations according to their requirements and do not have to spend much time coding numerical algorithms, particularly for in complex geometries. As an example, by using the ‘Plasma’ module, low-temperature plasma sources and systems (direct current discharges, inductively coupled plasmas, and microwave plasmas) can be simulated and scientists can gain insight into the physics of discharges and gauge the performance of existing or future designs. The only drawback of the solver is the lack of kinetic information. In fact, *COMSOL* does not include the Boltzmann equation solver and in many applications kinetic effects must be considered.

ANSYS Fluent [84] is a computational fluid dynamics (CFD) software tool used by researchers and engineers to optimize a product’s performance. *ANSYS Fluent* includes well-validated physical modeling capabilities to deliver fast and accurate results across the widest range of CFD and Multiphysics applications. *ANSYS Fluent* offers many great advantages in the modeling of plasmas. In fact, *ANSYS Fluent* can solve very complex fluid dynamic problems by using a large set of numerical algorithms to solve energy, momentum, mass, and many other kinds of

equations. Moreover, chemical reactions, continuity of species, drift–diffusion models, and Maxwell equations can be coupled and solved by taking advantage of a very important capability of such software: the user-defined scalar (UDS) variables. *ANSYS Fluent* can solve the transport equation for an arbitrary UDS in the same way that it solves the transport equation for a scalar such as species mass fraction. Extra scalar transport equations may be needed in certain types of combustion applications or, for example, in plasma applications. *ANSYS Fluent* allows one to define additional scalar transport equations in a model.

By using this capability, for an arbitrary scalar Φ_k , *ANSYS Fluent* solves the equation

$$\frac{\partial \rho \Phi_k}{\partial t} + \frac{\partial}{\partial x_i} \left(\rho u_i \Phi_k - \Gamma_k \frac{\partial \Phi_k}{\partial x_i} \right) = S_{\Phi_k} \quad k = 1, \dots, N, \quad (1.47)$$

where Γ_k and S_{Φ_k} are the diffusion coefficient and source term supplied by the user for each of the N scalar equations, u_i represents the velocity field, and x_i the spatial coordinates.

In the case of a steady-state problem and in the absence of convective fluxes $\frac{\partial \rho \Phi_k}{\partial t} = 0$ and $u_i = 0$. Furthermore, imposing $\Gamma_k = 0$, equation (1.47) becomes

$$-\nabla^2 \Phi_k = S_{\Phi_k} \quad k = 1, \dots, N. \quad (1.48)$$

ANSYS Fluent can solve UDS equations in both the fluid and solid regions and is able to impose a continuity condition along a boundary that separates a fluid zone from a solid one. This capability allows it to solve a large number of equations.

Both *COMSOL* and *ANSYS Fluent* are CFD codes based on the theoretical general models described in detail in chapters 6 and 7, and are applied in inductively coupled plasma torches (chapter 10) and aerospace applications (chapter 12).

BOLSIG+ [85] is a non-commercial code for the numerical solution of the Boltzmann equation for free electrons in weakly ionized gases in the framework of the two-term approximation (see chapter 2). Developed by Hagelaar [86] of the LAPLACE laboratory in Toulouse in France, *BOLSIG+* was first released in 2005 as a replacement for the earlier *BOLSIG* solver developed by Pitchford, Boeuf, and Morgan, based on the old code *ELENDIF* by Morgan and Penetrante [87].

BOLSIG+ uses data from the web-accessed database LXcat [49], a selection of complete sets of electron–neutral cross-section sets (see section 1.2) for a wide range of gases, compiled by different contributors and with varying levels of detail.

The main application of *BOLSIG+* is to obtain electron transport coefficients and rates for electron-induced processes starting from cross-section data. These quantities, which can be used as input for fluid models (see chapters 6, 7, and 12), are obtained from the non-Maxwellian electron energy distribution calculated for a given gas composition and under stationary conditions.¹¹

¹¹ This procedure, known also as cold gas approximation, should be compared with the self-consistent approach (see chapter 8).

BOLSIG+ is solved under the following assumptions:

- The electric field and collision probabilities are constant in space and time and there are no boundaries for the system.
- A constant gas composition.
- The electron production/loss due to ionization/attachment is assumed to result in exponential growth/decay of the electron number density.

Under the above assumptions, the Boltzmann equation reduces to a convection–diffusion equation with non-local source term in energy space, which is then discretized by an exponential scheme and solved for the electron energy distribution function by a standard matrix inversion technique (see section 2.7).

There are other non-commercial codes. These codes have been developed by research groups and generalized for a category of plasma applications. Typically these codes can simulate only 1D or 2D problems for a limited range of plasma applications. Among them, one of the best and most popular codes is *nonPDPSIM* [88], developed by Kushner and his research group. *nonPDPSIM* is a 2D computational platform that solves transport equations for charged and neutral species, Poisson’s equation for the electric potential, the electron energy conservation equation for the electron temperature, and radiation transport and Navier–Stokes equations for the neutral gas flow. The 2D platform has three major modules addressing plasma dynamics, fluid dynamics, and surface kinetics, and is implemented on an unstructured mesh to enable resolution of a large dynamic range in space. *CFD-GEOM* or *SKYMESH2*, a mesh generation program, has been incorporated into the program to allow for unstructured meshes. The model is coupled with a thermodynamics module providing local thermodynamic equilibrium properties. To speed up the simulation, a fully implicit scheme and sparse matrix techniques are used. The source function for electron impact reactions and secondary electron emission, and the electron transport coefficients needed by this segment of the code, are computed assuming the local field approximation, solving the electron energy equation and/or using a Monte Carlo simulation. The continuity equations for neutral species are solved explicitly.

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